

World Health Organization

Life Expectancy in *Developing* and *Developed* countries

Group assignment *Advanced R For Life Sciences*

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Introduction

Life expectancy

How to have a nice, long life?

What factors influence life expectancy?

Life expectancy: the number of years an individual can expect to live, on average
Calculated by mortality rates

Differences developed and developing countries?

How to efficiently increase life expectancy?

Introduction

WHO Data

Data from the World Health Organisation on life expectancy

For 179 countries from 2000 to 2015

18 variables:

- Biology:
  - Vaccination and disease
  - Mortality
  - BMI and child thinness
- Social:
  - Alchohol consumption
  - Schooling
- Economic:
  - Developing / Developed
  - GDP

Introduction

WHO Data

Developed vs developing:

- World Trade Organisation data
- Countries classify themselves

Note! *Vaccination* grade, not incidence for:

- Hepatitis B
- Measles
- Polio
- Diphteria

Data exploration

Validity

Missing / weird values?

First dataset wrong

Year	Population Afghanistan
2000	293.756
2001	2.966.463
2003	21.979.923
2004	2.364.851

Fixed dataset:

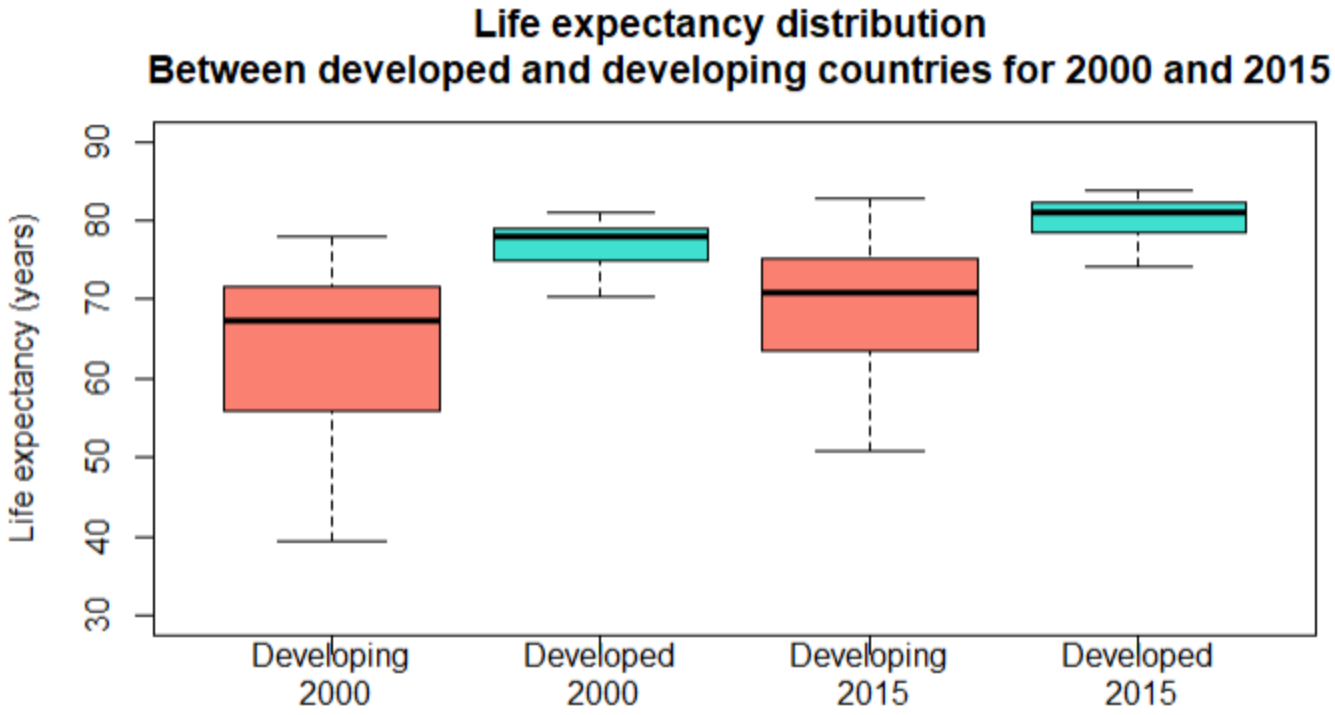
Missing values filled: 3 year or region

No NA, no missing values!

All 179 countries present for all 16 years

Spoiler: we found weird data

Introduction



Introduction

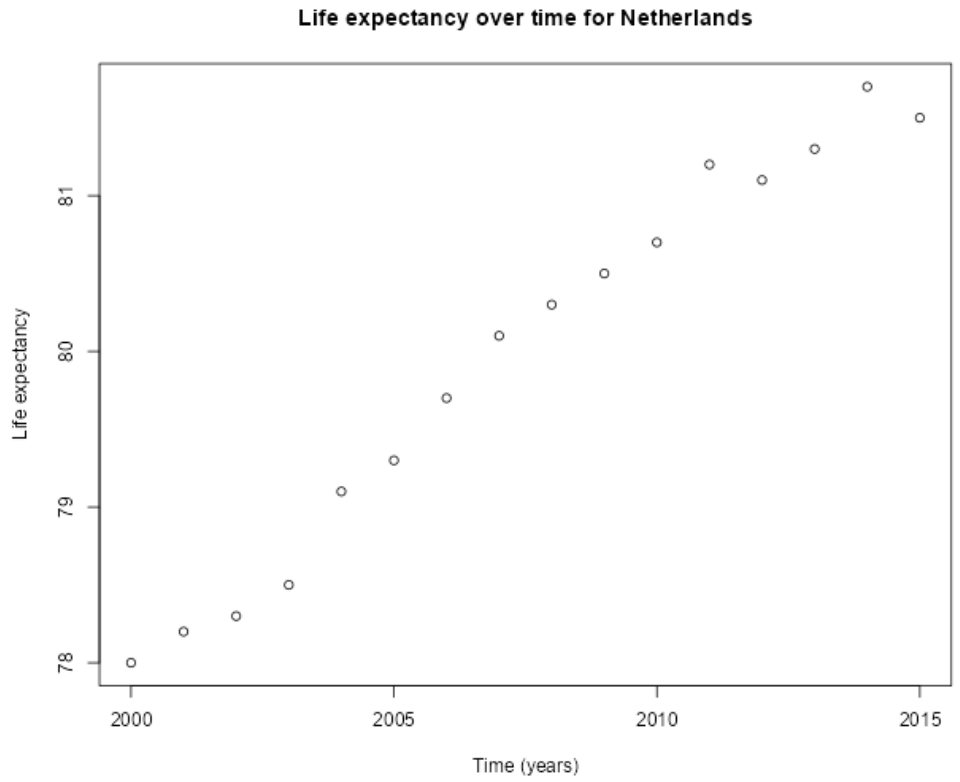
Responsive life extectancy graph

This graph allows you to select any country in the dataset and then plot its life expectancy over time

Select a country

Netherlands

Life expectancy over time plot



Multiple regression analysis

- Investigate which factors are associated with life expectancy
- 16 potential explanatory variables
- $R^2 = 0.979$ and $p < 0.001$
- No significant assiociation found for **Population size** ($p = 0.50229$)**, Measles vaccinations** ($p = 0.41835$)**, Thinness (age 5-9)** ($p = 0.87845$)**, Polio vaccinations** ($p = 0.69404$) **and Diphtheria vaccinations** ($p = 0.88476$)

Multiple regression analysis

Variable	Association	P value
Schooling	+	$p < 0.001$
Infant deaths	-	$p < 0.001$
Under-five deaths	-	$p < 0.001$
GDP per capita	+	$p < 0.001$
Hepatitis B vaccinations	-	$p = 0.00267$
Thinness 10 – 19	-	$p = 0.04962$
Adult mortality	-	$p < 0.001$
HIV incidents	+	$p < 0.001$
BMI	-	$p < 0.001$
Alchohol consumption	+	$p < 0.001$
Developed economy	+	$p < 0.001$

Multiple regression analysis

- Standardized model
- Scale()
- Mean = 0 and Standard deviation = 1
- Comparison of standardized coefficients (β)

Variable	Standardized β
Adult mortality	-0.590
Under five deaths	-0.266
Infant deaths	-0.140
GDP per capita	+0.046
BMI	-0.034
Schooling	+0.032
Alchohol consumption	+0.027
Developed economy	+ 0.028
HIV incidence	+ 0.028
Thinness age 10-19	-0.016
Heppatitis B vaccinations	-0.013

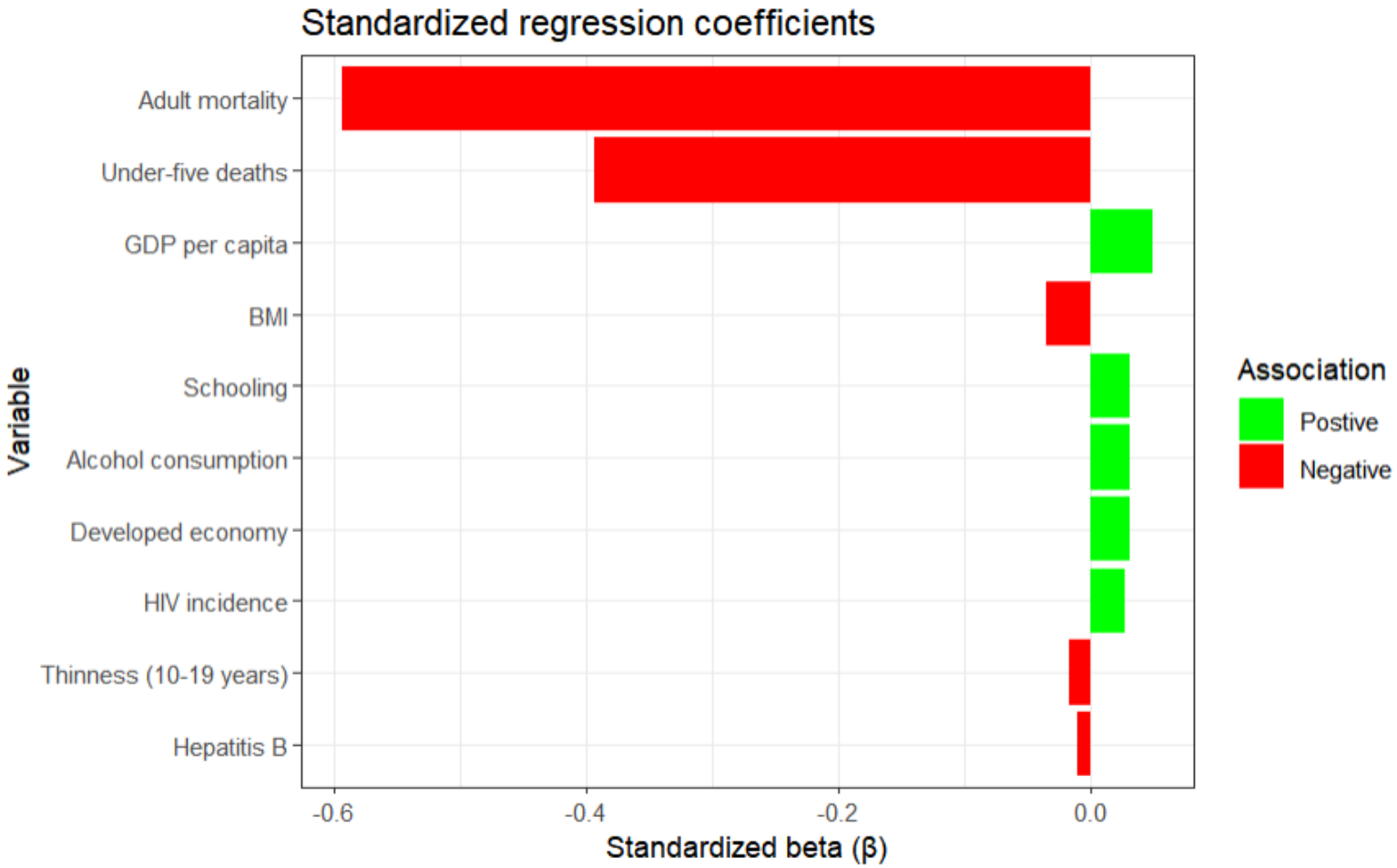
Multiple regression analysis

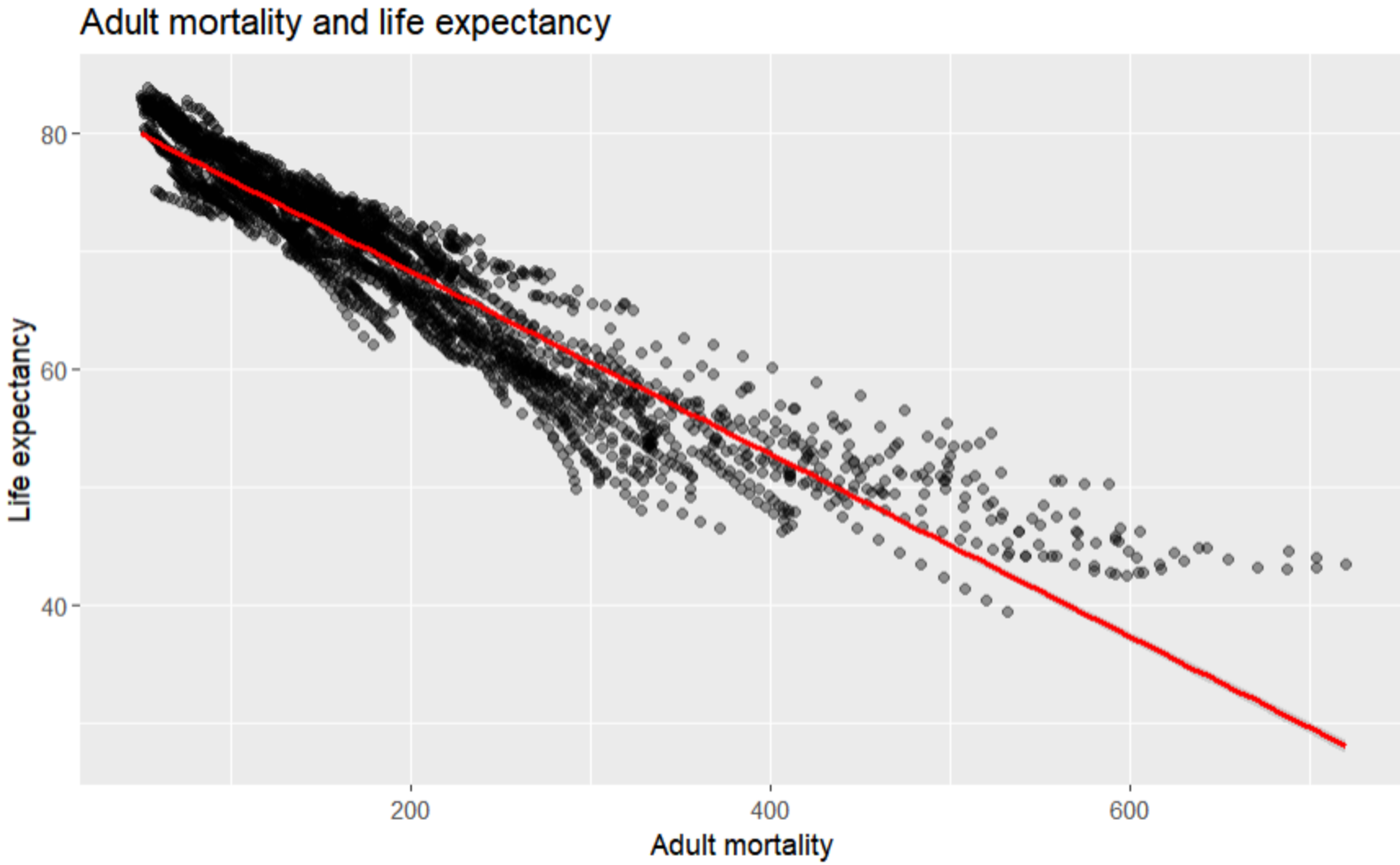
- **Checking for multicollinearity**
- Variance Inflation Factor (VIF)

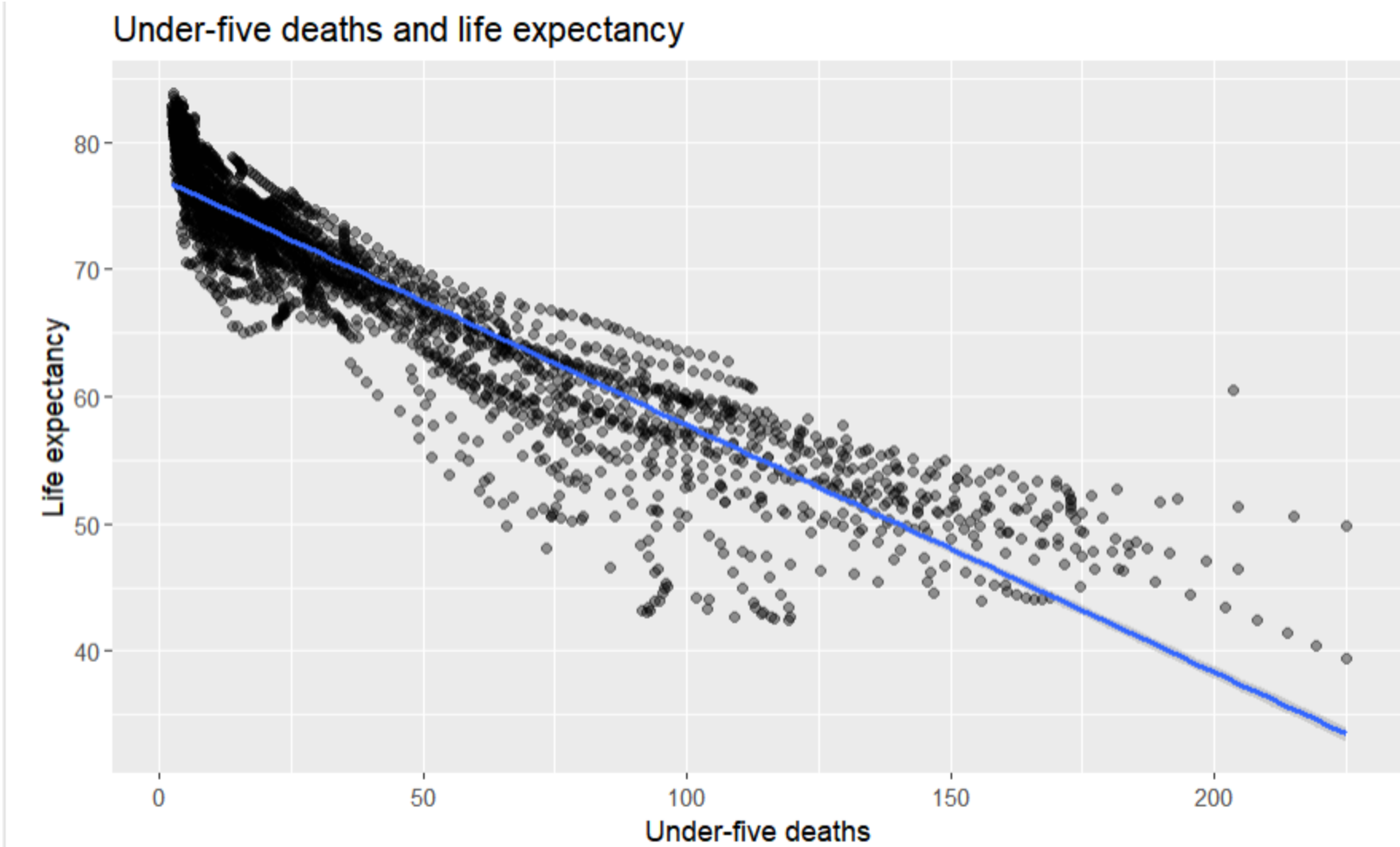
Variable	VIF	VIF in reduced model
Infant deaths	44.76	
Under five deaths	45.29	7.93
Polio	12.01	
Diphtheria	13.00	4.02
Thinness age 5-9	8.96	
Thinness age 10-19	8.95	2.00

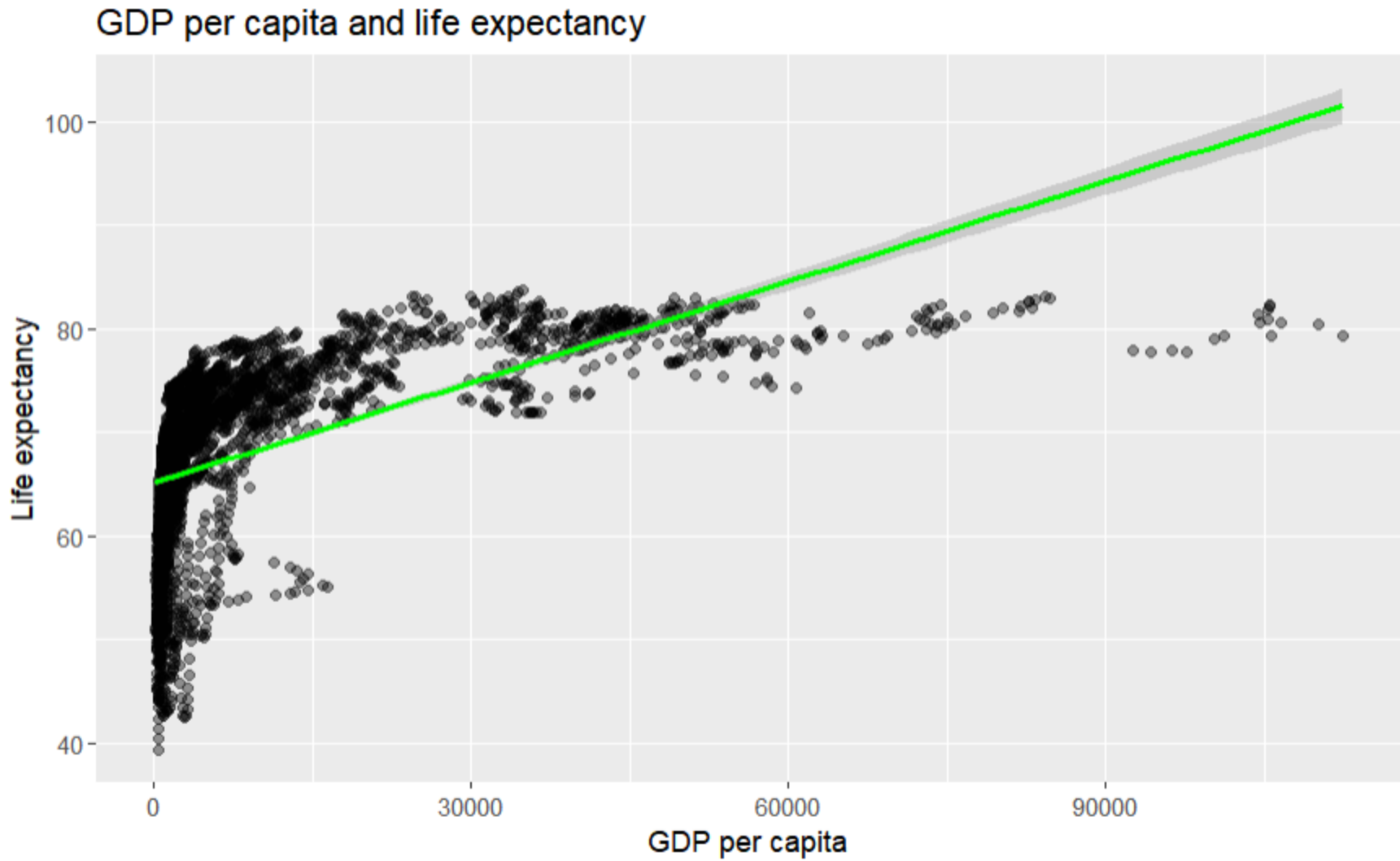
Multiple regression analysis

- **Final standardized model** (R^2 = 0.9787, p < 0.001)




Adult mortality and life expectancy
Life expectancy
80
60
40
200
400
600
Adult mortality

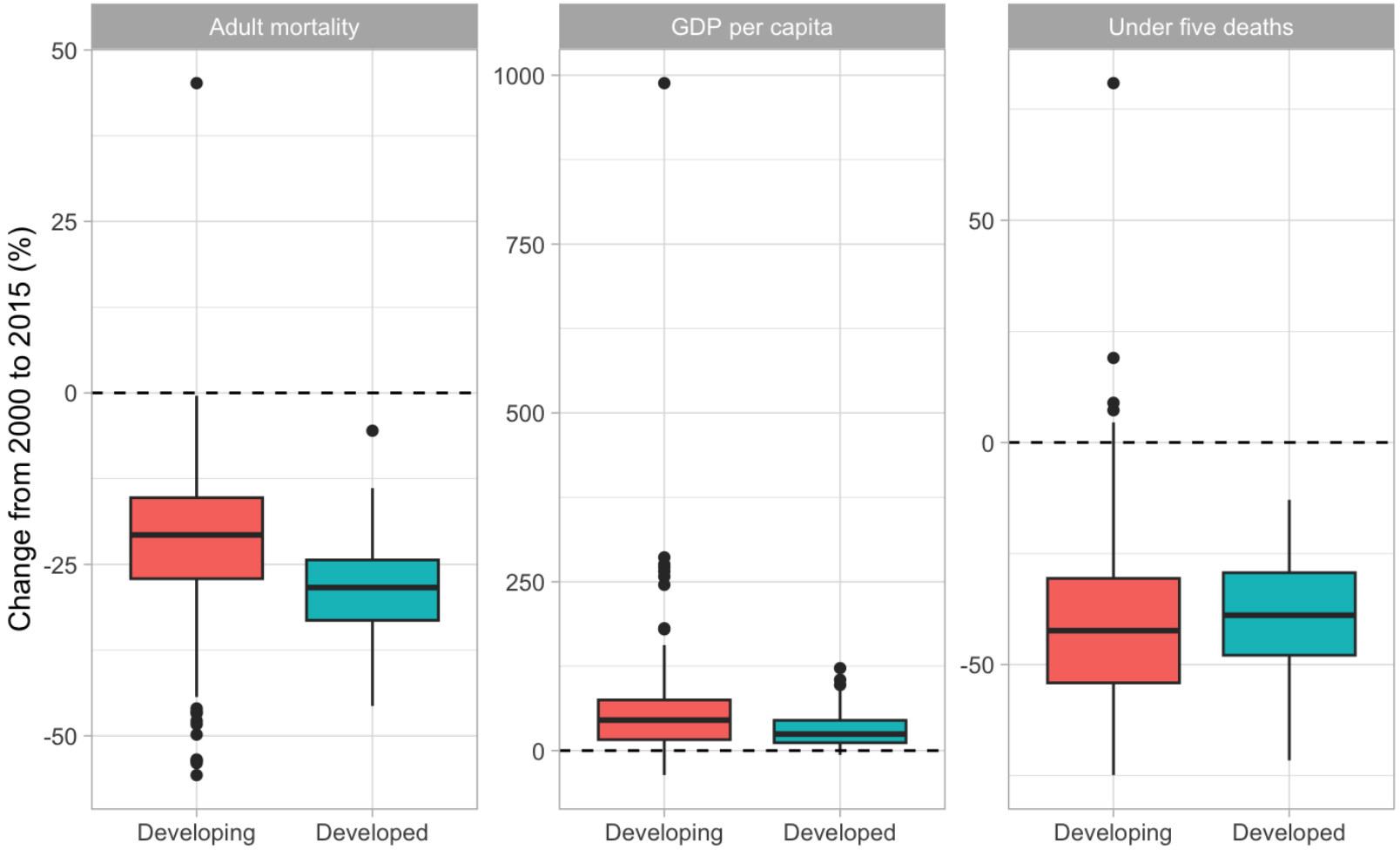

Under-five deaths and life expectancy
Life expectancy
80
70
60
50
40
0
50
100
150
200
Under-five deaths


GDP per capita and life expectancy
Life expectancy
100
80
60
40
0
30000
60000
90000
GDP per capita

Developed vs Developing countries

- **Boxplot overview**

Three determinants with the strongest correlation to life expectancy

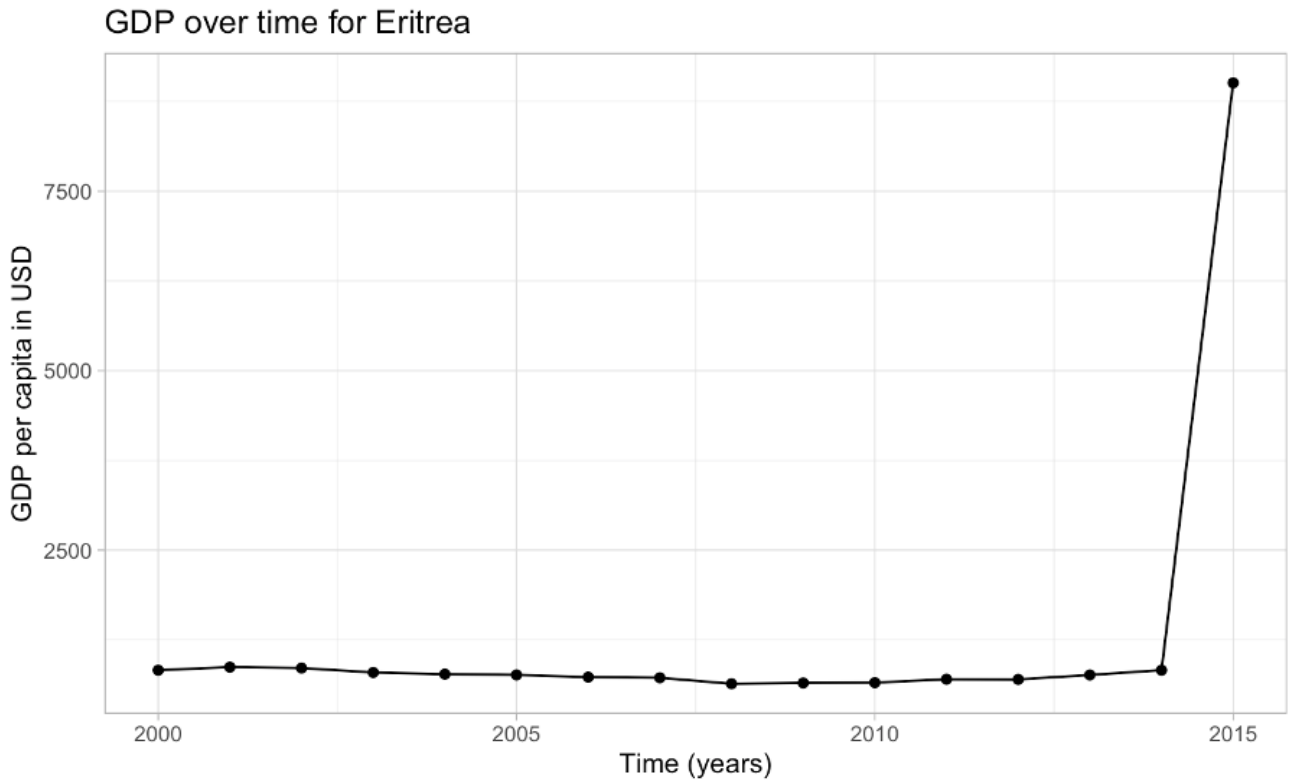


Developed vs Developing countries

❑ Boxplot overview

Three determinants with the strongest correlation to life expectancy

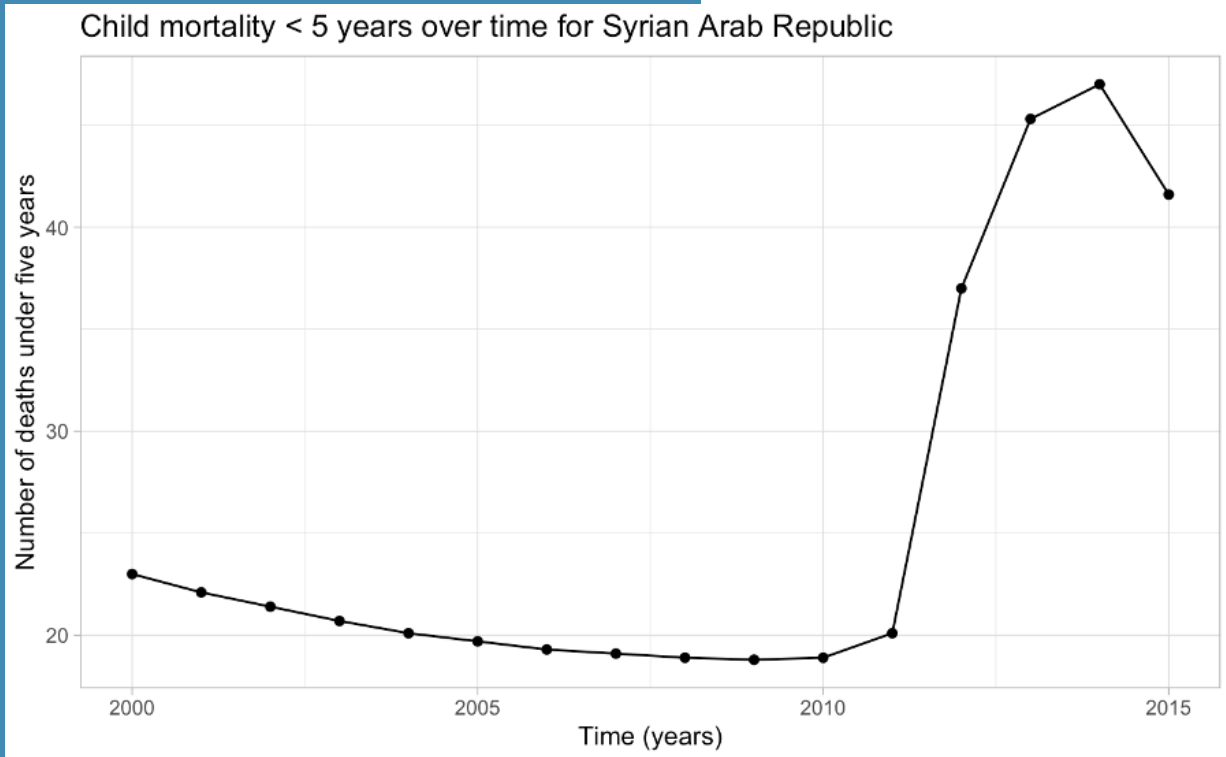
*Outlier in GDP data: Eritrea

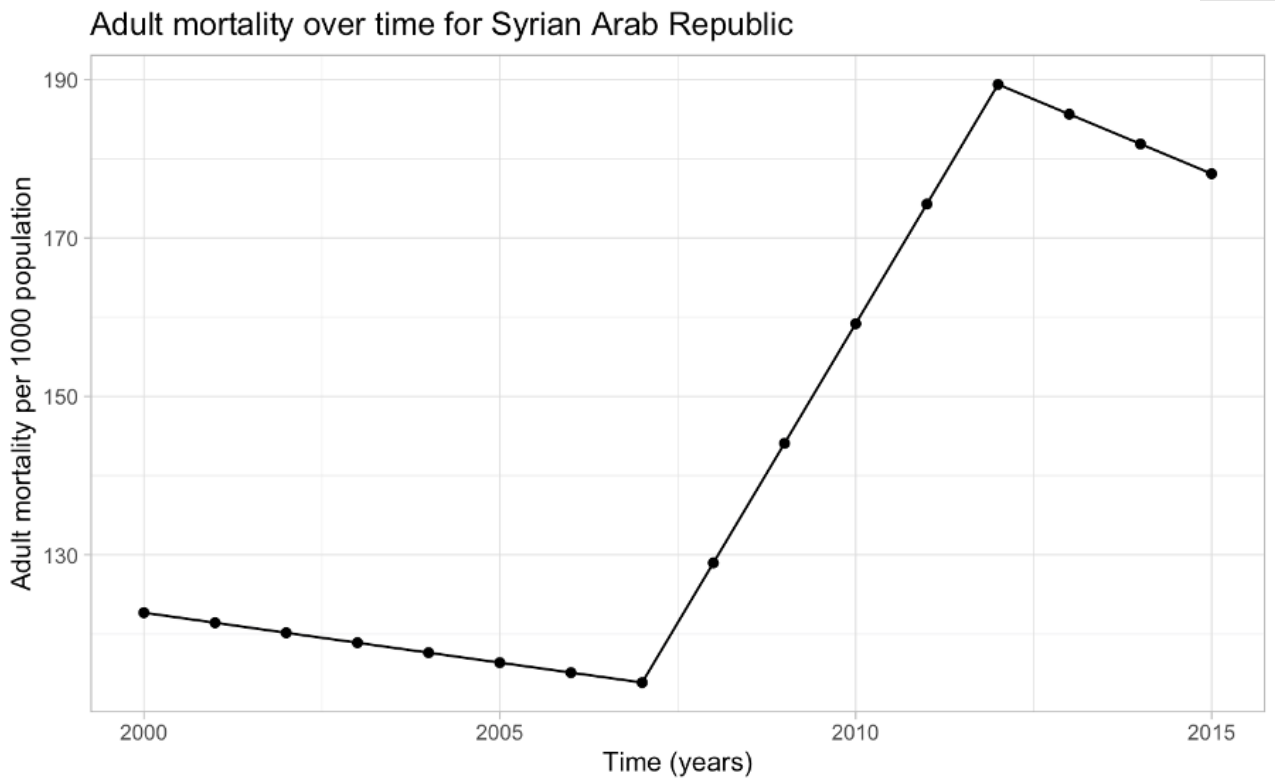


❑ Boxplot overview

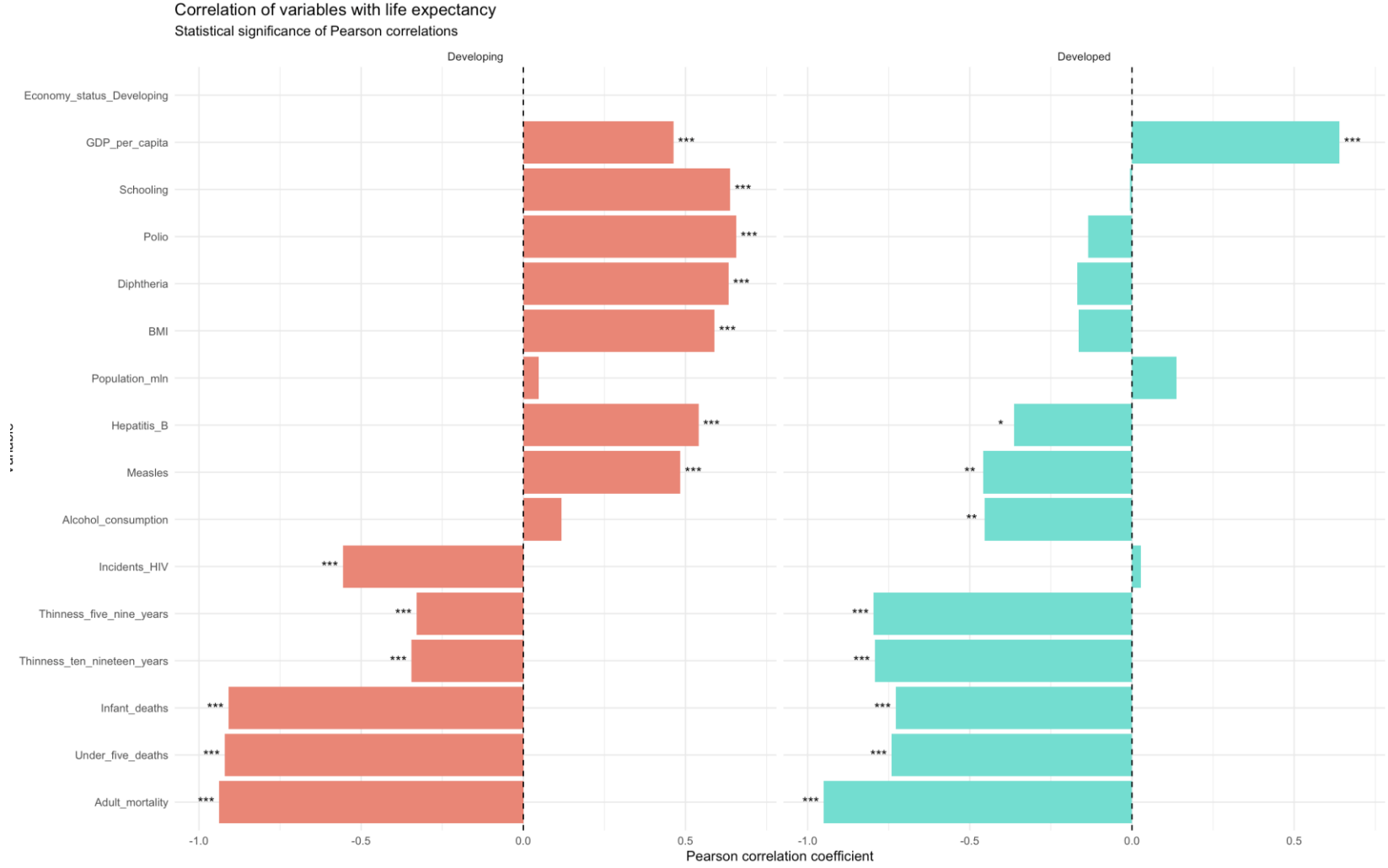
Three determinants with the strongest correlation to life expectancy

Syria outliers Adult mortality & Under five mortality

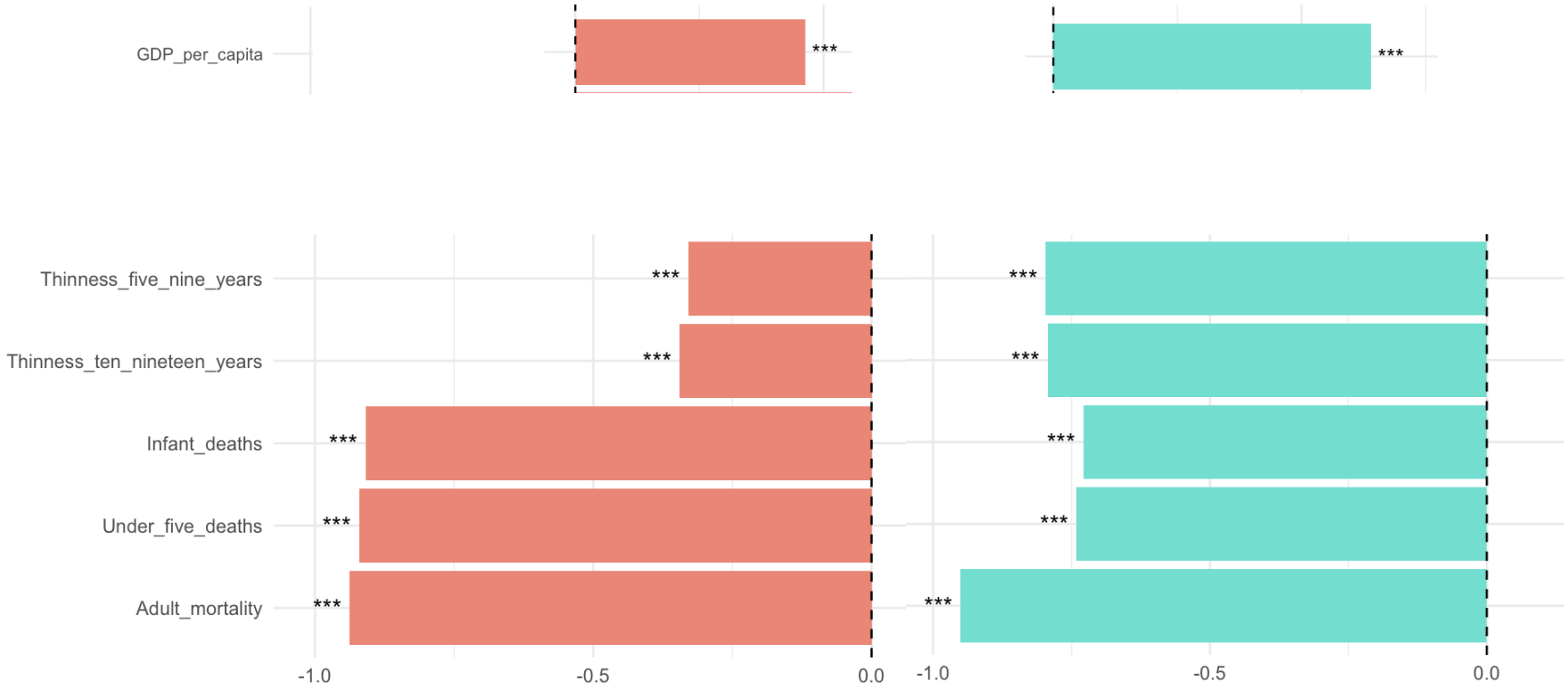




Correlations in Developing and Developed



Correlations in Developing and Developed



Correlations in Developing and Developed



Shiny

Life Expectancy Data

Data selection

Select year:

2000 2012 2015

2000 2002 2004 2006 2008 2010 2012 2014 2015

Select indicator:

Life_expectancy

Colour scatterplot by:

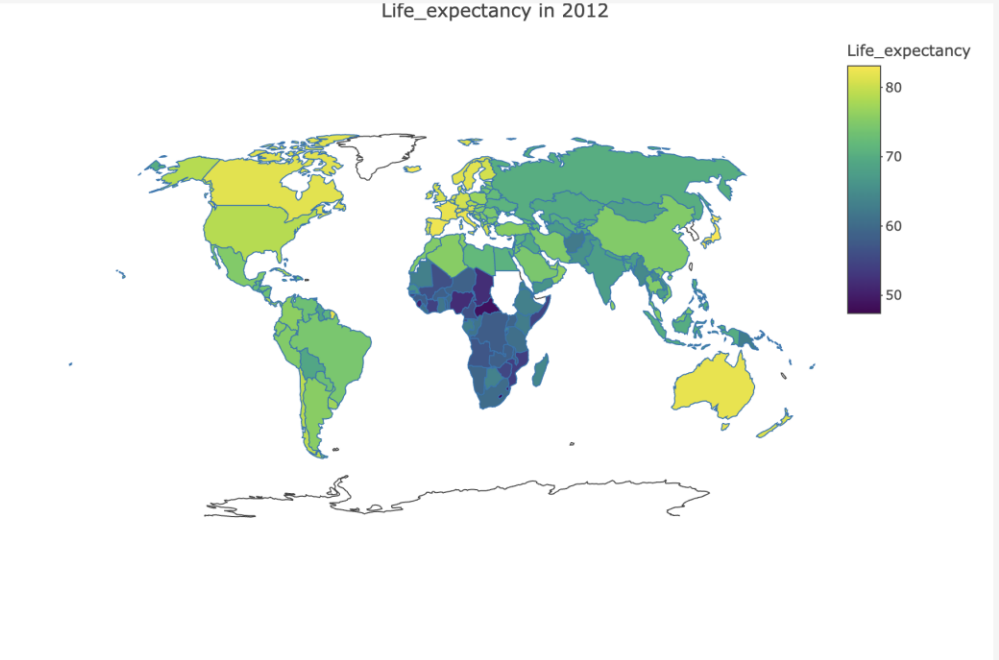
Development status

Minimum population size (mln):

0.08 100

0.08 10 20 30 40 50 60 70 80 90 100

Global distribution



Scatter plot



Conclusion

- Data inconsistent
  - big jumps in data points between years
- Data does not explain everything
  - Example: direction vaccinations developed vs developing
  - Underlying factors; socio-economic etc
  - Less children, linked to less child related deaths too
- How to live long? Don't die!